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Research on the detection of academic text plagiarism among college students using deep learning

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Abstract

Plagiarism of academic texts among college students is common, and there is an urgent need for an efficient detection method to strengthen the management of academic texts among college students. This paper proposes a Bidirectional Encoder Representations from Transformers-Bidirectional Long Short-Term Memory (BERT-BiLSTM)-attention matching model based on deep learning to determine whether sentence pairs match, thereby detecting plagiarism. Word vectors were obtained through BERT, and deep features were extracted from texts based on BiLSTM-attention. The Softmax function outputted the results. Experiments found that the BERT-BiLSTM-attention model achieved a F1 value of 87.56% for the Microsoft Research Paraphrase Corpus (MSRP) set, outperforming methods such as term frequency-inverse document frequency, and a Plagdet of 82.38% for the PAN12 set, thus outperforming other methods in the literature being compared. The model achieved an F1 value of 91.4% for real academic texts. The results demonstrate the excellent performance of the BERT-BiLSTM-attention model in detecting text plagiarism behavior, demonstrating its practical applicability.

Keywords Deep learning, College student, Plagiarism behavior, Text matching

1 Introduction

Influenced by the popularity of computers and the development of network technology, digital resources have become increasingly common and accessible [1], and the openness and convenience of the network have also led to more and more copying and plagiarism of such resources [2]. In the face of this situation, the detection of plagiarism has become crucial for the protection of intellectual property rights and social order [3]. Traditional methods, which rely on manual detection, are not only resource-intensive but also prone to subjectivity. Under the influence of improved computer performance, deep learning methods have been widely applied in natural language processing (NLP) [4]. Wang et al. [5] used an improved TextRank algorithm for word-level filtering of key sentences, learned features based on Bidirectional Encoder Representations from Transformers (BERT), and determined the matching relationship of long texts through



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a gate mechanism, achieving better outcomes than the baseline model on the CNSE and PAN datasets. Jia et al. [6] developed a text matching model of mixed multi-granularity language features and experimentally found that the Hit@1, Hit@3, and Hit@10 of the method were 0.821, 0.953, and 0.993, respectively. Cao [7] designed a text matching-based financial services robot chat system using an attention-convolutional neural network (CNN) structure. The experiment found that the method achieved 80.1% for accuracy and 82.5% for F1 score, respectively. In recent years, plagiarism among college students has been on the rise [8]. For college administrators, plagiarism not only hinders the cultivation of students with independent and innovative thinking but also greatly undermines the quality of higher education. Therefore, it is necessary to detect plagiarism among college students' academic texts. In terms of plagiarism detection, Hemandkumar et al. [9] put forward a machine learning-based approach for plagiarism detection in online assignments and achieved relatively high accuracy. Sharmila et al. [10] proposed Fuzzy Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) Language Modeling to detect plagiarism in decision science, and experiments on benchmark datasets found that the method was effective. This article designed a BERT-Bidirectional Long Short-Term Memory (BiLSTM)-attention detection model based on deep learning. Validation on the dataset verified that this method can be used as a tool for plagiarism detection in colleges to create a positive academic environment. In the first section of this paper, the designed BERT-BiLSTM-attention model is introduced. In the second section, experiments are carried out on the designed model, and the performance of this model in text plagiarism detection is verified. In the third section, the research results are briefly summarized, and conclusions are drawn.

2 A matching model based on deep learning

2.1 Text plagiarism behavior detection

The detection of plagiarism can be regarded as a semantic match of text [11], essentially measuring the semantic similarity between two texts. One of the current plagiarism detection methods is called paraphrase plagiarism detection [12], which determines whether there is plagiarism by analyzing the paraphrase relationship between the suspicious document and the source document. Generally, it includes the following two steps.

- (1) For a given collection of source documents D and a suspicious document d_{plg} , suspicious plagiarized documents D_{src} are identified from D . D_{src} contains the plagiarized source of d_{plg} .
- (2) Based on the semantic matching model, each d_{src} in $d_{\text{src}} \subseteq D_{\text{src}}$ is compared with d_{plg} to find out all plagiarized sections.

2.2 BERT-BiLSTM-attention matching model

In traditional text matching, the more identical words appear in two texts, the more similar the two texts are. The advent of deep learning has greatly improved the performance of text matching. With deep learning, text data can be represented in a digital way, i.e., words in the text are represented as word vectors, and then the matching degree of two texts is calculated. Based on deep learning, this paper designs a BERT-BiLSTM-attention matching model. After vectorizing a text with BERT, the semantic representation of the text is obtained through BiLSTM-attention. After vector concatenation, the matching

degree of two texts is obtained through value mapping, achieving the detection of plagiarism in academic texts. The basic structure of the model is displayed in Fig. 1.

(1) Word embedding layer.

The BERT pre-training model uses a bidirectional Transformer structure (Fig. 2). Compared with traditional methods such as word-to-vector (Word2vec), BERT can focus on all the contexts on both sides of a word simultaneously, thereby automatically learning richer semantic knowledge, and thus performing well in NLP tasks [13]. According to Fig. 1, when processing text, BERT adds the word vector, sentence vector, and position vector to obtain $E_1, E_2, \dots, E_N$. After bidirectional Transformer, word vector representation $T_1, T_2, \dots, T_N$ is output.

(2) BiLSTM-attention layer.

Long short-term memory network (LSTM) is an optimization of a recurrent neural network (RNN) [14], effectively alleviating the problems of vanishing or exploding gradients that exist in RNNs. Therefore, it demonstrates outstanding performance in scenarios such as data prediction and text processing [15]. The calculation process of LSTM is as follows.

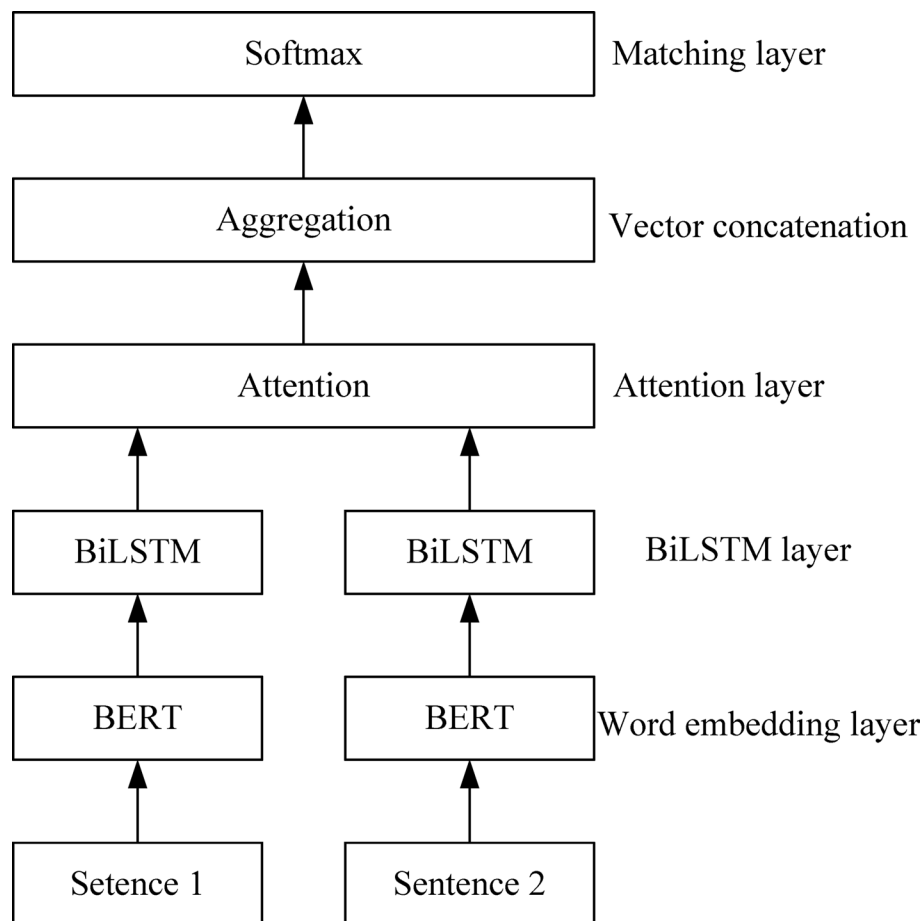


Fig. 1 BERT-BiLSTM-attention matching model

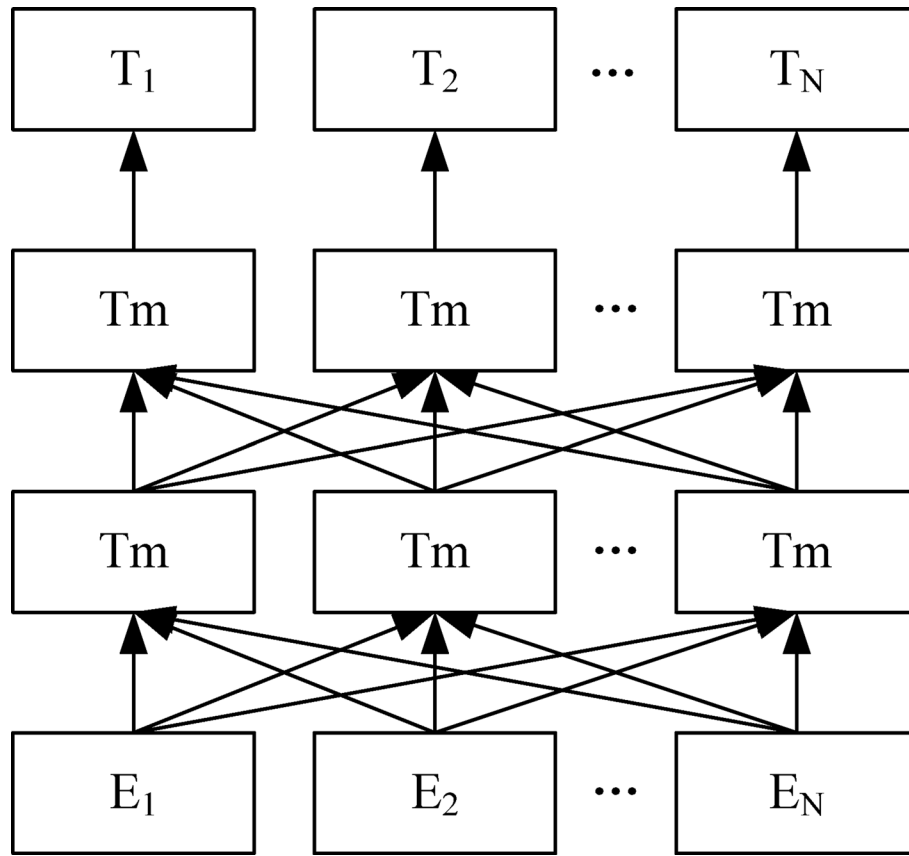


Fig. 2 BERT structure

It is determined by the input gate i_t how much of the current input will be sent to the cell state:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i),$$

where h_{t-1} refers to the hidden state of the previous cell, x_t is the current input, and W and b are the weight and threshold respectively. Then, forgetting gate f_t determines how much of the previous cell state will be sent to the current cell state:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f).$$

Then, based on the tanh function, a new candidate state is established:

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c).$$

The cell state is updated: $c_t = f_t c_{t-1} + i_t \tilde{c}_t$.

The output gate controls how much of the current cell state will be sent to the hidden state:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o).$$

Ultimately, the hidden state is obtained: $h_t = o_t * \tanh c_t$.

However, LSTM can only encode the information from front to back and has insufficient learning of the information from back to front. Therefore, BiLSTM [16], which combines forward and backward LSTMs, can capture bidirectional semantic

Table 1 Examples of the MSRP dataset

Example sentence	Tag
A: The new Finder puts a user's folders, hard drive, network servers, iDisk and removable media in one location, providing one-click access	1
B: Panther's redesigned Finder navigation tool puts a user's favourite folders, hard drive, network servers, iDisk and removable media in one location	
C: The product also features an updated release of the Apache Web server, as well as Apache Tomcat and Apache Axis	0
D: Panther Server also includes an updated release of Apache, along with Apache Tomcat and Apache Axis for creating powerful web services	

Table 2 MSRP dataset information

Training sentence pair	2000
Testing sentence pair	1000
Positive example	1500
Negative example	1500

dependencies. The output of BiLSTM is the result of concatenating the outputs of forward and backward LSTMs.

The role of the attention layer is to extract key semantic features from the BiLSTM output to achieve attention to important information [17], and the calculation process is:

$$\alpha = \text{softmax}(w^T M),$$

$$M = \tanh(H),$$

$$r = H\alpha^T,$$

where α is the attention weight and r is the final BiLSTM-attention layer output. The output vectors of sentence 1 and sentence 2 are concatenated, and the softmax function is applied in the matching layer to obtain the matching result. An output of 0 indicates that the input sentence pairs do not match, while 1 indicates a match.

3 Results and analysis

3.1 Experimental dataset

- (1) Microsoft Research Paraphrase Corpus (MSRP) dataset [18]: it is commonly used for text similarity tasks, where the texts are derived from original news on the web. It contains a total of 5,801 pairs of text, with a label of 1 for semantically matched text pairs and 0 for vice versa. The example sentences of the dataset are presented in Table 1.

To ensure data balance, some data were extracted from the MSRP dataset for experiments (Table 2).

The evaluation of the MSRP dataset adopted the text classification method:

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN},$$

Table 3 The PAN12 dataset2

Obfuscation strategy	Instruction	Number of training sentence pair	Number of testing sentence pairs
No plagiarism	No plagiarism	1000	500
No obfuscation	Copy and paste directly	1000	500
Low obfuscation	Simple text obfuscation	1000	500
High obfuscation	Substantial text obfuscation	1000	500
Paraphrasing	Manual rewriting	1000	500
Translation	Translation confusion	1000	500

Table 4 Parameter setting

Parameter	Value
Word vector dimension	200
Hidden layer dimension	128
Optimizer	Adam
Learning rate	0.001
Epoch	100
Dropout	0.5

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall},$$

where TP is the number of true positive cases, FP is the number of false positive cases, and FN is the number of false negative cases.

- (2) PAN12 dataset [19]: The data comes from books in Project Gutenberg. The creator used a series of obfuscation strategies to establish plagiarized text pairs. The dataset information is displayed in Table 3.

The plagiarism detection indicators used in the PAN12 dataset are as follows.

It is assumed that S is the collection of plagiarized texts in the PAN12 dataset, $s = s_{plg}, d_{plg}, s_{src}, d_{src}, s \subseteq S$, and R is the collection of plagiarized texts detected by an algorithm, $r = s_{plg}, d_{plg}, s_{src}, d_{src}, r \subseteq R$. Then,

$$Precision(S, R) = \frac{|\bigcup_{(s,r) \in (S \times R)} (s \cap r)|}{|\bigcup_{r \in R} r|},$$

$$Recall(S, R) = \frac{|\bigcup_{(s,r) \in (S \times R)} (s \cap r)|}{|\bigcup_{s \in S} s|},$$

$$Granularity(S, R) = \frac{1}{|S_R|} \sum_{s \in S_R} |R_s|,$$

where S_R is the plagiarized text detected in S and R_s is the number of intersections between the plagiarized text and the detection results.

Finally, the above three subtables are combined to calculate Plagdet:

$$Plagdet(S, R) = \frac{F1}{\log_2(1 + Gran(S, R))}.$$

3.2 Experimental setup

The experiment environment was the Windows 10 operating system, and the algorithm was implemented using the Keras framework of TensorFlow. The BERT pre-training model provided by the hugging-face platform was used, and the other parameters were set as shown in Table 4.

3.3 Result analysis

Using the MSRP dataset, the effects of different modules in the BERT-BiLSTM-attention model were determined using ablation experiments. Moreover, it was compared with the most advanced model: Transformer. The experiment was repeated five times.

From Table 5, it can be seen that compared with the BERT-BiLSTM-attention model, the BiLSTM model had significantly lower precision, recall rate, and F1 value. By comparing the BERT-BiLSTM and BiLSTM-attention models with the BERT-BiLSTM-attention model, it was found that removing either the BERT module or the attention module could lead to a significant decline in the model's performance, which verified the reliability of both modules. By comparing the Transformer model with the BERT-BiLSTM-attention model, it can be found that the latter achieved better results for the MSRP dataset ($p < 0.05$). Overall, the BERT-BiLSTM-attention model significantly outperformed its simplified versions and the most advanced model in terms of all evaluation indicators. The BERT-BiLSTM-attention model achieved depth extraction of text features through a cascaded architecture of BERT (deep semantic initialization), BiLSTM (context modeling), and attention (focus on key points). Compared with using BERT alone, BiLSTM alone, or simple combinations (BERT-BiLSTM and BiLSTM-attention), it can capture the core semantics of text more accurately. It has better and more stable performance in plagiarized text detection. The hybrid architecture of BERT-BiLSTM-attention is also more effective than the pure Transformer, which verified the reliability of this model.

Moreover, the BERT-BiLSTM-attention method was compared with other baseline methods, including:

- (1) The term frequency-inverse document frequency (TF-IDF) method [20]
- (2) Arc-II: A convolutional neural network-based method [21]
- (3) The multi-granularity convolutional network (MultiGranCNN) method [22].

According to Fig. 3, the TD-IDF method performed poorly for the MSRP dataset, with an F1 value of only 77.62% and a low precision of only 70.31%, indicating a high false alarm rate. This is because this method was merely a statistic of word frequency and lacks understanding of deep semantics. The Arc-ii, MultiGranCNN, and BERT-BiLSTM-attention methods are all deep learning-based methods that performed significantly better than the TF-IDF method, demonstrating the superiority of deep learning methods in text matching. Specifically, the Arc-II method achieved a recall rate of 96.04% for the MSRP dataset, but with a precision of only 69.91%. Although the method had a low tolerance for missed detections, it only achieved an F1 value of 80.92%, showing poor matching performance. The MultiGranCNN method achieved a relatively better balance between precision and recall rate compared to the TF-IDF and ARC-II methods, with an F1 value of 87.56%, indicating that the MultiGranCNN method captured

Table 5 Ablation experiments

	Precision	Recall rate	F1
BiLSTM	79.85 ± 0.42*	88.76 ± 0.38*	84.07 ± 0.36*
BERT-BiLSTM	80.92 ± 0.35*	91.45 ± 0.41*	85.86 ± 0.33*
BiLSTM-attention	81.34 ± 0.38*	92.18 ± 0.36*	86.40 ± 0.31*
Transformer	81.77 ± 0.35*	92.66 ± 0.33*	87.01 ± 0.31*
BERT-BiLSTM-attention	82.23 ± 0.31	93.62 ± 0.29	87.56 ± 0.28

$p < 0.05$ compared to the BERT-BiLSTM-attention model

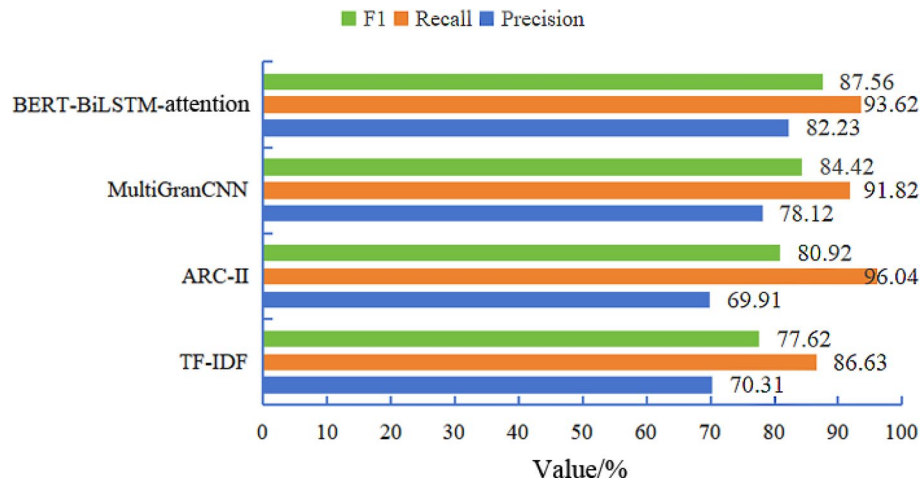


Fig. 3 Results for the MSRP dataset

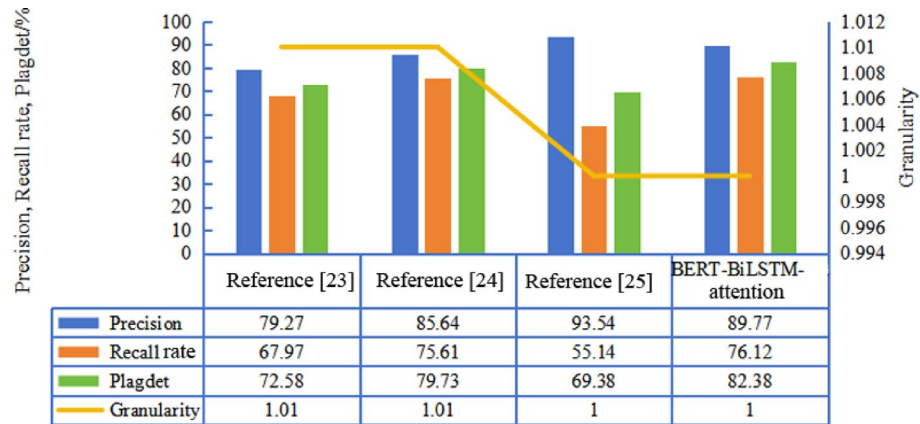


Fig. 4 Results for the PAN12 dataset

more features and thus performed better in text matching. Finally, the BERT-BiLSTM-attention method achieved an F1 value of 87.56%, which can effectively capture the long-term dependencies of texts and better determine whether texts match or not. The results demonstrate the effectiveness of the proposed method.

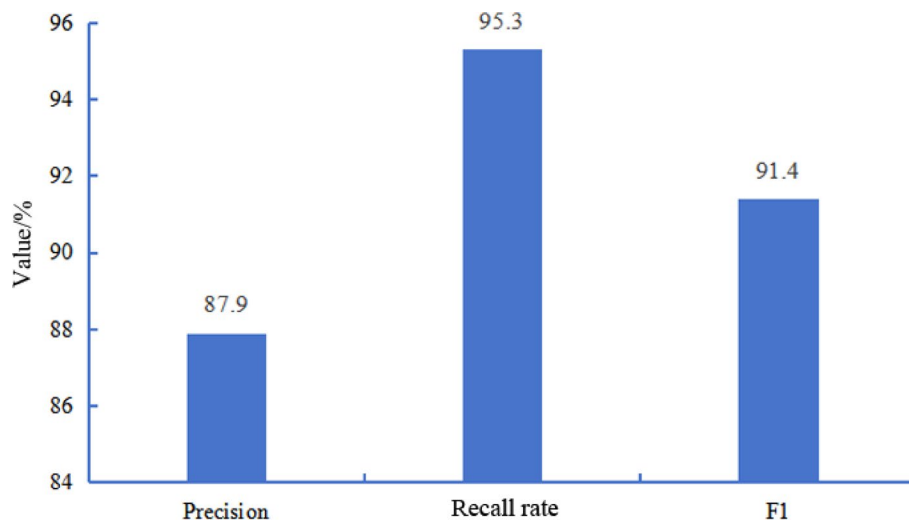
Using the PAN12 dataset, this method was compared with other plagiarism detection methods, including:

- (1) the method based on cosine distance and Jaccard coefficient [23];
- (2) the semantic-syntactic similarity measurement method based on WordNet [24];
- (3) the LSTM-based method [25].

According to Fig. 4, the recall rate of the method based on cosine distance and Jaccard coefficient for the PAN12 dataset was relatively low, at only 67.97%, indicating that it may not be sensitive to some plagiarism behaviors and has some missed detections. The Plagdet was also relatively low, at only 72.58%, suggesting that its performance in detecting text plagiarism behaviors was average. The WordNet method achieved a better balance between precision and recall rate, with a Plagdet of 79.73%, indicating that it can detect more plagiarized texts while maintaining good accuracy. The LSTM-based method achieved the highest precision at 93.54%, but its recall rate was extremely low at

Table 6 Confusion matrix

	Detected as plagiarism	Detected as no plagiarism
Plagiarism actually	TP = 102	FN = 5
No plagiarism actually	FP = 14	TN = 379

**Fig. 5** The detection performance of the BERT-BiLSTM-attention model

55.14%, indicating a large number of missed detections when detecting plagiarized texts, and its Plagdet was only 69.38%. Although the precision of the BERT-BiLSTM-attention method was only 89.77%, not as good as that of the LSTM-based method, its recall rate was 76.12%. There was a good balance among the three indicators, and the final Plagdet was 82.38%. The findings highlight the reliability of the BERT-BiLSTM-attention method in detecting text plagiarism.

Moreover, the proposed method was applied to real academic text detection. Among 500 academic texts, 107 texts were marked by the teachers as having plagiarized content. The results obtained by detecting using the BERT-BiLSTM-attention model are displayed in Table 6 and Fig. 5.

It can be found from Table 6 and Fig. 5 that the BERT-BiLSTM-attention model detected most cases of plagiarism. Among the 107 texts with plagiarism behavior, the model successfully detected 102 texts, showing good reliability. The F1 score reached 91.4%. This result indicates that the model can serve as an efficient preliminary screening tool, providing reliable assistance for the manual review of academic text plagiarism in universities. Thus, it can reduce the scope of manual re-checking and better safeguard academic integrity.

4 Discussion

The advancement of information technology has transformed the environment of academic creation. The Internet provides convenient materials and tools for college students' academic creation and also provides conditions for academic misconduct, especially text plagiarism. Traditional plagiarism detection, which relies on rule-based methods such as string matching and keyword comparison, performs poorly in detecting text plagiarism involving semantics, with prominent cases of missed and false

detections; therefore, it is difficult to deal with the increasingly serious situation of academic text plagiarism among college students. The wide application of methods such as deep learning in NLP tasks provides a brand-new tool for plagiarism detection. It is necessary to study the usability of deep learning in plagiarism detection due to the need to address new types of plagiarism and improve the efficiency of detection.

This paper combines the BiLSTM algorithm with BERT and attention to design a text matching model. The architectural combination of BERT, BiLSTM, and attention is not an original idea. However, the research in this paper is not simply a stacking of architectures. Instead, it is a framework for detecting academic text plagiarism, which enables the collaborative work of BERT, BiLSTM, and attention. In the specific scenario of academic text plagiarism detection, BERT has deficiencies in capturing long-distance logical consistency across paragraphs and chapters. The combination of BERT and BiLSTM can effectively solve this problem and capture the argumentation logic and structural features spanning multiple sentences in academic texts. The introduction of attention can focus more on key content, such as core argument statements and unique terms. This paper conducted experiments on the MSRP and PAN12 datasets. According to the results of the ablation experiment on MSRP, it can be found that each module is useful in this task, which verifies the necessity of this combination in the plagiarism detection task. According to the comparison results, it can be found that the designed matching model showed excellent performance in capturing semantic features of the two datasets. It achieved a precision of 82.23% and an F1 value of 87.56% for the MSRP dataset and a precision of 89.77% and an F1 value of 82.38% for the PAN12 dataset, indicating that the BERT-BiLSTM-attention model performed well in text matching. It can detect plagiarized texts accurately, with few missed and false detections. Plagiarism detection on real students' academic texts showed that the designed model can detect plagiarism relatively accurately. Therefore, it can be applied in detecting academic text plagiarism among college students in actual universities.

This study provides a new and applicable method for detecting text plagiarism. For college administrators, it offers a more intelligent academic supervision tool to deter plagiarism and purify the academic atmosphere. For students, it is conducive to cultivating their rigorous academic attitude and promoting their academic growth, thereby strengthening talent cultivation.

However, the research in this paper also has some limitations. For instance, the proposed method only targets the detection of plagiarism within the same language. In the current real-world situation, there are many cases of plagiarism across different languages. Moreover, only text features are utilized, while in actual academic texts, there is also information such as images and code. In addition, the means of plagiarism in academic texts among college students are becoming increasingly diverse. The model developed in this paper does not take into account the situation where plagiarism involves using large language models (LLMs) to generate and rewrite content. Therefore, in future work, the detection method for plagiarism will be further improved to achieve the detection of cross-language plagiarism. More multi-modal features will be considered to enhance detection performance. Experiments will be conducted on a wider range of datasets, such as those datasets containing plagiarism generated and rewritten by LLMs to verify the usability of the model.

5 Conclusion

While the advancement of artificial intelligence empowers education, it also provides new means for academic plagiarism. This paper used deep learning to design a BERT-BiLSTM-attention model for detecting plagiarism in college students' academic texts. Experiments on public benchmarks (MSRP and PAN12) showed that the model can effectively capture the deep semantic associations in plagiarized texts, and its detection performance is better than that of multiple baseline models, demonstrating the advantages of deep learning methods in this task. The research in this paper validates the practicality of deep learning models in plagiarism detection tasks. It provides an effective artificial intelligence solution for universities to prevent and control increasingly intelligent academic plagiarism, and also offers an available baseline model for subsequent research to address more complex plagiarism scenarios, such as those involving rewriting generated by LLMs.

Author contribution

Shen Jiang designed research, performed research, analyzed data, and wrote the paper.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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